

VISHVESH TRIVEDI

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US Permanent Resident (Green Card) — No Visa Sponsorship required

Education

New York University, Courant Institute of Mathematical Sciences

Sep 2024 – May 2026

Master of Science, Computer Science,

GPA: 4.0/4.0

National Institute of Technology, Surat

Dec 2020 – May 2024

Bachelor of Technology, Computer Science,

GPA: 9.07/10.0

Coursework: Data Structures & Algorithms, Programming Languages, Deep Learning, CV, NLP, Operating Systems, Database Management, Computer Architecture, Networks, Probability & Statistics, Mathematics, GPU

Experience

Biomedical Data Sciences Hub, NYU Langone Health

Nov 2024 – Present

Data Engineering Intern

New York, United States

- Led query optimization and SQL plan enhancement for large-scale OMOP Common Data Model (CDM) tables on an Azure Databricks (Apache Spark) cluster, significantly improving execution efficiency for analytics workloads.
- Designed an end-to-end ETL pipeline to transform EHR records from Caboodle to OMOP CDM in real time. Experience with EPIC COSMOS EHR data (300M+ records), HIPAA compliance and SQL performance optimization.
- Contributed to NIH and PCORi grant proposal submissions (\$12M in total funding) by generating performance insights of clinical AI tools and providing visualizations on pilot data used by research and operations teams.

Center for Visual Information Technology, IIIT Hyderabad

Jan 2024 – Aug 2024

Software Engineer Intern - AI Research

Hyderabad, India

- Engineered parallelized multimodal data generation pipelines (Python, HPC, SLURM) for large-scale synthetic data generation with auto-annotation for downstream tasks thereby reducing high-quality data procuring cost by over 75%.
- Trained a backend-retrieval engine over CLIP embeddings and deployed within a microservice architecture, to support large-scale multimodal search across slide images and text resulting in an 8% improvement in Recall@K.
- Published findings as an oral presentation at ICDAR 2025. Open-sourced code and dataset with Apache 2.0 License. Received 2000+ impressions and 500+ downloads within a month of release on HuggingFace.

Wells Fargo

May 2023 – Jul 2023

Software Development Engineer Intern

Hyderabad, India

- Pioneered a web-based fullstack tool using Next.js and Django that produces semantic-aware audio-transcriptions of PPT presentations that is 40% faster than screen-readers, and directly impacts 15000 visually impaired WF employees.
- Designed an end-to-end CI/CD pipeline and data version control manager for efficient deployment workflows. Leveraged Docker and Jenkins-based automation to reduce redeployment latency by 25%.
- Spearheaded the team showcase event and completed 4 professional certifications on ML-Ops best practices, Data Governance, Agile, and Scrum methodologies. Received full-time return offer but turned down for higher studies.

Projects

Adaptive and Warp-Cooperative GPU Hash Table [\[Code\]](#)

CUDA, C++, Bash, Linux, GPU, Data structure Internals, Parallel Algorithms, High Performance Computing

- Built a high-performance adaptive GPU hash table in C++/CUDA, implementing cooperative groups and elected-lane atomics, achieving 21× faster inserts, 20× faster lookups and outperforming naïve GPU hashing at scale.
- Implemented epoch-based dynamic resizing + compaction, enabling non-blocking concurrency and sustaining stable throughput on 100M+ operations even at 0.99 load factor.

RepCRec: A lightweight implementation of a fault-tolerant, distributed database. [\[Code\]](#)

C++, Database Internals, Fault Tolerance, Distributed Algorithms, Replication and Sharding

- Built a distributed database supporting Serializable Snapshot Isolation, transaction scheduling, and failure recovery.
- Developed test harness and simulation framework to validate correctness over dozens of distributed transaction scenarios ranging up to 90% simulated site failures, 100,000 tx/sec data stream and 10× data replication factor.

Open Source contribution to Retrieval Heads project [\[Code\]](#)

vLLM, ZeRO, flash-attention, PyTorch, Python, Hugging Face Transformers, GitHub, Open Source

- Rewrote the entire codebase of Retrieval Heads (ICLR 2025 spotlight paper) to make it operate 2x faster
- Designed high-throughput dynamic dataloaders in Pytorch, vectorized all tensor operations, and used flash-attention library and vLLM framework for fast inference serving.

Technical Skills

Languages/DBs: Python, C++, JavaScript, Typescript, SQL, Bash, CUDA, PostgreSQL, MongoDB, MySQL, NoSQL

Frameworks: React, FastAPI, Node.js, Express.js, Python, LangChain, Django, Next.js, PyTorch, Hugging Face

Tools/Platforms: Docker, Kubernetes, Jenkins, Redis, Kafka, AWS (EC2, S3, IAM), GCP, Azure, PySpark, Databricks